claim that turned out to be right and that foretold the hairlessness of holes. I thus proceeded cautiously. “I had no idea that a spinning metal sphere emits electromagnetic radiation. How?”

“The radiation is so weak,” Zel’dovich explained, “that nobody has ever observed it, nor predicted it before. However, it must occur. The metal sphere will radiate when electromagnetic *vacuum fluctuations* tickle it. Similarly, a black hole will radiate when gravitational vacuum fluctuations graze its horizon.”

I was too dumb in 1971 to realize the deep significance of this remark, but several years later it would become clear. *All* previous theoretical studies of black holes had been based on Einstein’s general relativistic laws, and those studies were unequivocal: A black hole cannot radiate. However, we theorists knew that general relativity is only an approximation to the true laws of gravity—an approximation that should be excellent when dealing with black holes, we thought, but an approximation nonetheless.[7] The true laws, we were sure, must be quantum mechanical, so we called them the laws of *quantum gravity.* Although those quantum gravity laws were only vaguely understood at best, John Wheeler had deduced in the 1950s that they must entail *gravitational vacuum fluctuations*, tiny, unpredictable fluctuations in the curvature of spacetime, fluctuations that remain even when spacetime is completely empty of all matter and one tries to remove all gravitational waves from it, that is, when it is a perfect vacuum (Box 12.4). Zel’dovich was claiming to foresee, from his electromagnetic analogy, that these gravitational vacuum fluctuations would cause spinning black holes to radiate. “But how?” I asked, puzzled.

Zel’dovich bounded to his feet, strode to a one-meter-square blackboard on the wall opposite his map, and began drawing a sketch and talking at the same time. His sketch (Figure 12.1) showed a wave flowing toward a spinning object, skimming around its surface for a while, and then flowing away. The wave might be electromagnetic and the spinning body a metal sphere, Zel’dovich explained, or the wave might be gravitational and the body a black hole.

Box 12.4

Vacuum Fluctuations